Department of CSE, GEC Idukki
BTech Computer Science and Engineering (2025-2029)
Odd Semester 2026-27
S3 CSE - PCCSL 307 Data Structure Lab Cycle

Exp.No	Experiment	CO mapping	Completion Date
	Cycle 1		
1	Insert or delete a given element from an array.	CO1	22/7/26
2	Read, store, and display the details (roll no, name, CGPA) of n students in a class using array of structures.	CO1	22/7/26
3	Find the sum of two polynomials using array of structures.	CO1	22/7/26
4	Find the transpose of a sparse matrix using array of structures.	CO1	28/7/26
5	Find the sum of two sparse matrices using array of structures.	CO1	28/7/26
6	Implement a singly linked list. Write functions to (i) Create a list (ii) Insert a node (after and before a given node value; get user choice) (iii) Delete a node with the given value (iv) Display the list	CO2	28/7/26
7	Implement addition of polynomials using singly linked lists.	CO2	4/8/26
8	Implement a doubly linked list.	CO2	4/8/26
9	Simulate a browser history using a doubly linked list. Each node represents a visited web page. The program should perform the following operations: (i) Visit a new web page. (ii) Move to the previous page (Back). (iii) Move to the next page (Forward). (iv) Display the current page. (v) Display the complete browsing history.	CO2	4/8/26
10	Implement stack operations using arrays.	CO1	6/8/26
11	Implement stack using linked lists.	CO2	6/8/26
12	Convert an infix expression to postfix expression and then evaluate it. (using stacks).	CO1	6/8/26
13	Implement queue operations using arrays.	CO1	6/8/26
14	Implement DEQUEUE and Circular Queue using arrays.	CO1	6/8/26